

Social Engineering and Phishing

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An emergency call to a father

South Korea, reported in July 2018

A caller pretended to be his daughter.

She claimed an accident and asked for ₩20 million urgently.

What would make it difficult to pause?

The daughter was already at home

The real daughter heard the call.

She spoke to the caller, who hung up.

No payment occurred in this account.



Which security check could work from another city?

https://www.korea.kr/news/policyNewsView.do?newsId=148852565

Week 2 still gives us the framework

A	**Asset**	Savings, account access and personal information.
V	**Weakness**	Sending money before independently checking the emergency
C	**Control**	Checking with the real daughter through a trusted route

By the end of today, you should be able to explain:



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- Human factors and social-engineering techniques
- Phishing, spear phishing, smishing, vishing and impersonation
- Warning signs and safe reporting procedures

1. Human Factors

Social-engineering techniques

What is Social Engineering

Deception that influences a person's action

An attacker uses a believable story to make someone ***reveal information****,* ***transfer money*** *or* ***bypass a safeguard****.*

The attacker might contact a person by phone, text, email, social media or in person. The important feature is the use of deception to influence action

Social Engineering and Human Factor

*Human factors are the ways **attention**, **emotion**, **habits** and the **working situation** affect decisions*

Social engineering is a cyberattack that specifically hacks the "human factor" rather than the computer code

Pretext and requested action

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Pretext

The situation the attacker invents.

Example: “Your scholarship payment failed.”

Requested action

What the attacker wants you to do.

Example: send a password or approve an unexpected login.

Why a believable request works

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T

Trust

We often cooperate with familiar people and institutions.

P

Pressure

Fear or a deadline can narrow our attention.

A

Attention

A busy moment can interrupt a normal checking habit.

Authority, fear and urgency

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A

Authority

- “I am from the court.”
- A title suggests power.

F

Fear

- “Your account is involved in a crime.”
- The story threatens harm.

U

Urgency

- “You have ten minutes.”
- The deadline discourages checking.

Familiarity, reward and curiosity

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F

Familiarity

- “It’s me. I changed my number.”
- A relationship creates trust.

R

Reward

- “You won a tuition grant.”
- The offer creates hope.

C

Curiosity

- “Someone posted your photo.”
- The message invites a quick look.

The harm can be personal

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M

Money

Losing savings may affect rent, tuition or family support.

I

Identity

An exposed account may reveal private messages or documents.

R

Relationships

An attacker can use a trusted account to approach friends.

Credentials and MFA

1	**Credentials**	Information used to sign in, such as a username and password.
2	**MFA**	Multi-factor authentication uses more than one type of proof.
3	**Approval prompt**	A request to approve a sign-in. Verify the login you started.

Uber: repeated login approvals

September 2022

A contractor initially rejected repeated sign-in requests.

Eventually, one approval allowed the attacker to log in.

What changed at the final approval?



The decision in the Uber case

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PASSWORD

1

The attacker had a contractor's password.

REQUESTS

2

Repeated approval prompts initially failed.

APPROVAL

3

One accepted prompt permitted access.

What is a more useful control than "be more careful"?

A claimed identity also needs a separate check

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Verified contact

- You reach the person or organization through a route you already trust.
- You also confirm the exact request.

Claimed identity

- The message says “university,” “bank,” “court” or “your friend.”
- Names, logos and profile pictures can be copied.

Baiting and tailgating

Social engineering can also happen face to face

Baiting

- An attractive item or offer invites a risky action.
- **Example:** an unknown USB labelled "**Exam answers.**"

Tailgating

- Someone follows an authorized person into a restricted area.
- **Example:** a visitor asks you to hold a secure door.

How can you be helpful without bypassing an access rule?

Conditions that support safer choices

P

People

Someone can pause and ask for help without embarrassment.

P

Process

Important requests have a known verification procedure.

T

Technology

Account controls limit what one mistaken action can expose.

Activity: the scholarship message

Fictional campus message
“You have been selected for a grant. Confirm your password and send the code now. Keep this offer private until approval.”

1. Name the pretext.
2. Identify two pressures.
3. Name the risky action.
4. Choose a separate check.

Story, action and check

S	**Story and pressure**	A grant offer uses reward, urgency and secrecy.
A	**Action and asset**	Password and code disclosure could expose the student account.
C	**Independent check**	Open the usual portal or contact the scholarship office separately.

2. Phishing and Impersonation

Channels, targeting and real cases

Phishing

Deceptive contact that seeks information or an unsafe action

A message pretends to be trustworthy and directs you to a ***request***, ***link***, ***attachment*** *or* ***conversation***.

Five terms describe different features

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Term	Feature	Meaning
Phishing	Deception	A broad category of deceptive contact
Spear phishing	Targeting	Personalized to a person, role or group
Smishing	Text channel	Phishing through SMS or text messages
Vishing	Voice channel	Phishing through a call or voice contact
Impersonation	Claimed identity	Pretending to be another person or organization

Broad phishing and spear phishing

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Broad phishing

- “Your account needs attention.”
- A general message goes to many people.

Spear phishing

- “Your Tuesday project group has a new assessment file.”
- Details match a particular class or person.

A message knows your project

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Fictional example

A sender uses your project title and your teammate's name. It asks you to open a new sign-in page to read "the instructor's feedback."

1. What makes it targeted?
2. What action could create harm?
3. Where should you check?

Smishing: a delivery text

Smishing = phishing through SMS or text

Fictional SMS
“Delivery failed. Update your address and pay a small fee through this link.”

Why might a small fee feel harmless?

Twilio: a text became an entry point

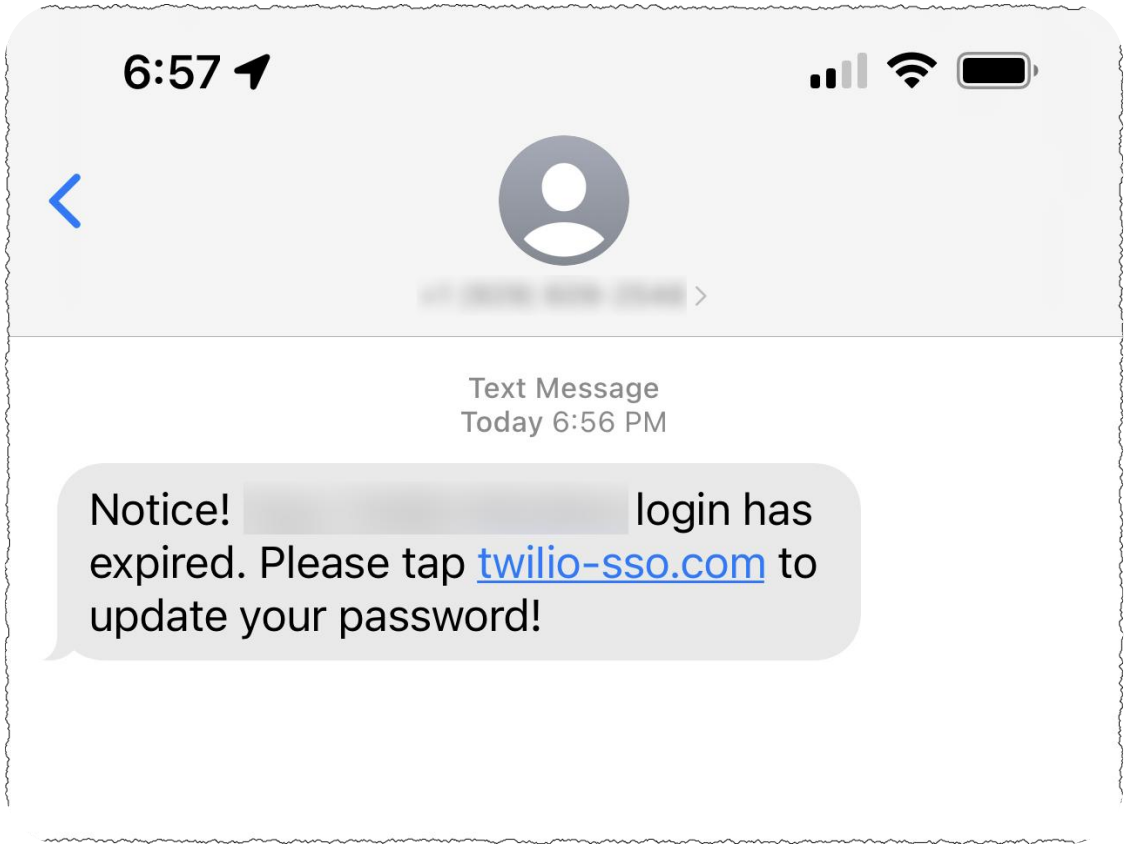
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August 2022, United States

Messages impersonated internal IT.

Some employees used fake sign-in pages, exposing credentials.

Does a familiar company name prove the destination?



Historical screenshot published by Twilio. Do not visit links.

Twilio: the risk extends to other people

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A	**Asset**	Customer information accessible through internal tools.
E	**Event**	Stolen employee credentials enabled unauthorized access.
C	**Control purpose**	Protect sign-in and limit the reach of a compromised account.

Vishing: a voice carries the request

V	**Voice channel**	A caller may claim to be support, a bank or a relative.
P	**Pressure**	Conversation can create urgency and discourage a pause.
A	**Action**	The caller asks for money, a login code or device access.

South Korea: a fake court delivery

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Official warning, Changwon District Court, 8 July 2025
Scammers claimed court mail had been returned. Calls or messages urged people to use a link, provide information or install an app.

1. Name the claimed authority.
2. Explain the feared consequence.
3. Identify the risky request.
4. Choose an official contact route.

Why unfamiliar legal language can make someone want to cooperate quickly?

Unfamiliar language adds pressure

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Terms to understand

- Court notice
- A message about a legal process.

- Verification code
- A code that can authorize access.

A student's safer response

- Pause and get help understanding the request.

- Use a known university office or official contact route before acting.

Impersonation crosses channels

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F

Family or friend

A familiar relationship can support an urgent money story.

S

Staff or support

A work role can make an account request seem routine.

O

Organization

A university, bank or court name can suggest authority.

Deepfakes and identity claims

A deepfake uses AI to imitate a person’s appearance or voice.

Seeing a face or hearing a voice may increase confidence.

A sensitive request still needs an independent check.

Hong Kong: a fabricated meeting

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EMAIL

1

A fake chief financial officer (CFO) requested a confidential discussion.

VIDEO

2

A fabricated, pre-recorded meeting appeared to confirm the request.

TRANSFER

3

The employee authorized transfers. Reported loss: about HK$200 million.

Authorization and recognition differ

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Authorization

- “This request passed our agreed verification and approval process.”
- A sensitive action needs more than recognition.

Recognition

- “That looks like my manager.”
- A familiar image can influence belief.

Activity: one label may be too few

Contact	Your labels and reason
A. Generic email requests a new sign-in.	
B. SMS uses your project name and a fake tutor identity.	
C. A caller posing as a bank asks for a code.	
D. A copied friend profile requests urgent money.	

Classification: answers with reasons

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Contact	Supported description
A. Generic email	Broad phishing. The story is general.
B. Tailored tutor SMS	Smishing + spear phishing + impersonation.
C. Fake bank call	Vishing + impersonation. Voice is the channel.
D. Copied friend profile	Impersonation through a message. The precise channel is unspecified.

The asset changes the consequence

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Personal situation

- An emergency story can target family savings.
- A student account can expose private information or interrupt study.

Workplace situation

- A compromised account can reach other people’s information.
- A payment instruction can expose organizational funds.

3. Warning Signs and Reporting

Warning signs need context

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U

Unexpected

A new contact, unusual login or sudden payment change.

P

Pressure

A deadline, threat or secrecy request blocks checking.

S

Sensitive action

The request concerns credentials, money or device access.

Requests that deserve a pause

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1	**Password or code**	Someone asks you to reveal a password or relay a login code.
2	**Payment change**	A contact introduces a new destination or urgent transfer.
3	**Device access**	A caller asks you to install an app or enable remote access.

A familiar word can hide a different site

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- https://login.campus.example/student
- https://campus.example.verify-now.test/student

The full hostname matters. Open the known portal yourself.

Appearance is limited evidence

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Stronger checks

- An expected task and a known contact route.
- Confirmation of the exact request through the official process.

Easy to imitate

- A logo, display name, fluent grammar or familiar layout.
- HTTPS can also protect a connection to a deceptive site.

Practice: an urgent tuition notice

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A QR code promises to prevent a tuition penalty if you sign in immediately.

What is the strongest next action?

A. Scan now **B. Open the usual portal** **C. Ask the sender**

Pause, verify and report

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PAUSE

1

Stop the requested action and end pressure from the contact.

VERIFY

2

Use a separate known route to confirm the exact request.

REPORT

3

Use the official channel and explain what happened.

A practical verification script

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On a call

- “I will contact the office through its official number.”
- End the call and find that route separately.

For a message

- Open the usual app or portal.
- Ask the known office: “Did you send this request, and is this action required?”

Reporting routes in South Korea

Situation	Route	Purpose
Money sent or active voice-phishing fraud	112 and your bank’s official contact	Report urgently and request payment suspension.
Suspicious SMS or possible malicious app	118 KISA(Korea Internet & Security Agency)	Get technical advice about the message or device.
Campus account or university message	Known university IT or responsible office	Report the message and arrange account response.

If you already acted

What happened	Immediate response
Opened a link only	Close it. Report and explain any downloads or permissions.
Entered credentials or approved a login	Use a trusted route to reset access and end sessions. Contact IT.
Installed an app or allowed remote access	Stop using the device for sensitive tasks. Seek help from another device.
Sent money	Contact 112 and the bank immediately. Request payment suspension.

A supportive response helps recovery

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Useful response

- “Thank you for telling me. What happened, and what did you already do?”
- Help the person reach the right reporting route.

Unhelpful response

- “How could you fall for that?”
- Shame can make someone hide information or delay help.

Q&A: four definitions

1. What does social engineering try to influence?

2. How does spear phishing differ from broad phishing?

3. What distinguishes smishing from vishing?

4. Why can impersonation overlap with the other labels?

Answers: the definitions

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1. It uses deception to influence a person's action.

2. Spear phishing single the contact to a person, role or group.

3. Smishing uses texts. Vishing uses voice contact.

4. Impersonation describes the false identity, which can appear in many channels.

Q&A: the decision matters

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1. Why do a correct name and logo fail to prove authenticity?

2. What makes a verification route independent?

3. What should change after someone shares a password?

4. How can reporting protect someone else?

Answers: evidence and response

1. A sender can copy details or use a compromised account.

2. You obtain the route separately from the suspicious contact.

3. Stop the interaction, secure the account through a trusted route and report the exposure.

4. A timely report can help block the attempt and warn other potential targets.

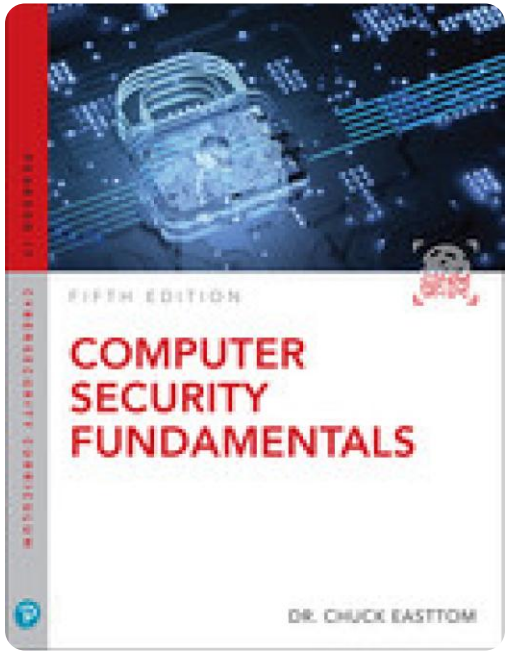
Assignment: one personal protection plan

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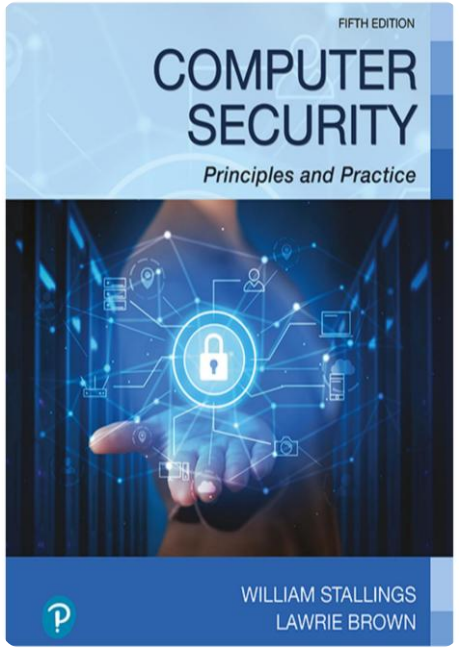
Write one page using a fictional student or one documented case.

Explain how you would protect an important account, payment or relationship.

1. Name the asset and threat.
2. Explain the manipulation.
3. Give a check and report route.



Chuck Easttom
Computer Security Fundamentals
5th edition



Stallings and Brown
Computer Security: Principles and Practice
5th edition

Sources for the real cases and response

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South Korean family account: Korea Policy Briefing, 23 July 2018.

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Uber: Security update, 19 September 2022.

Twilio: final incident report, 27 October 2022.

Hong Kong deepfake case: government reply, 26 June 2024.

Cloudflare: phishing incident report, 9 August 2022.

Safe response: Korean Financial Services Commission guidance.

Warning signs: U.S. Federal Trade Commission phishing advice.